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HYBRID ALPHA TRADING STRATEGY

COMBINING MULTIPLE ML MODEL FOR FORECASTING

PRESENTED BY : SLEEP NOT FOUND

MEET OUR TEAM



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**MACHINE LEARNING
ENGINEER**



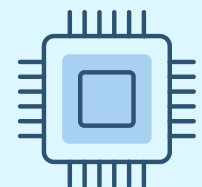
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**SYSTEM
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DESIGNER**

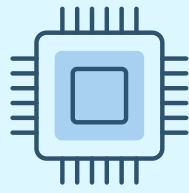


KU XIN QI

**FINANCE
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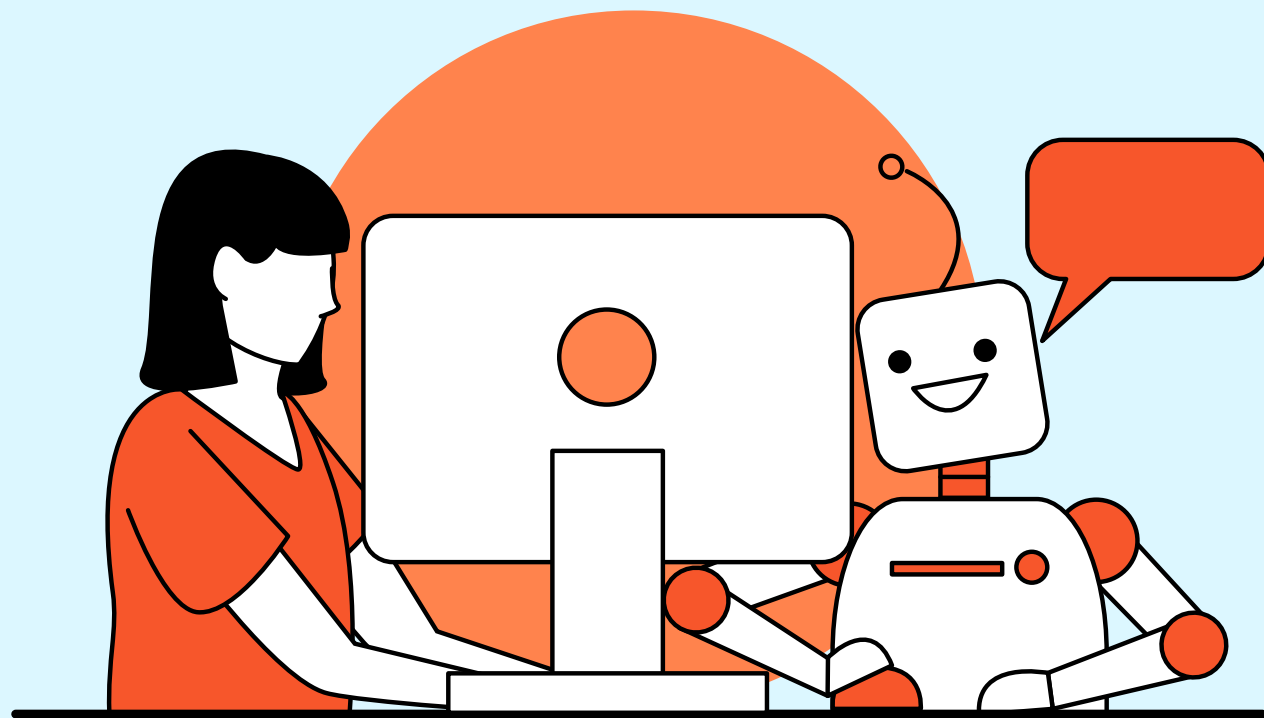


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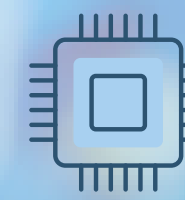


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CONTENTS



- Problem Statement
- Value Proposition
- Step-by-Step Methodology
- Trading Rules & Signal Generation
- Backtest Results
- Competitive Advantages of Alpha Trading
- What We Need



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PROBLEM STATEMENT

PROBLEM STATEMENT

Crypto Markets Are Highly Volatile and Non-stationary

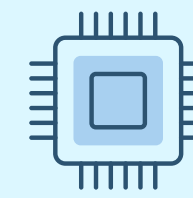
Cryptocurrencies experience rapid and unpredictable price movements, influenced by factors like global news, sentiment shifts, regulatory updates, and whale activity. These conditions make the market non-stationary

Traditional Indicators Struggle in Dynamic Environments

Technical indicators such as RSI, MACD, or moving averages are based on fixed time windows and assumptions. They lack the flexibility to adjust in real time and often generate misleading signals when market behavior shifts quickly.

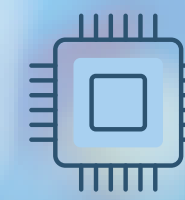
Most Model Ignore Regime Shifts and Context

Markets go through different regimes—bull, bear, accumulation, distribution—but many models treat all market data the same. This leads to inaccurate predictions, as they miss the hidden context behind the price movements.



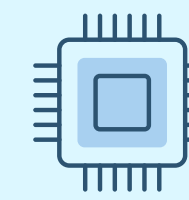
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OUR VALUE PROPOSITION

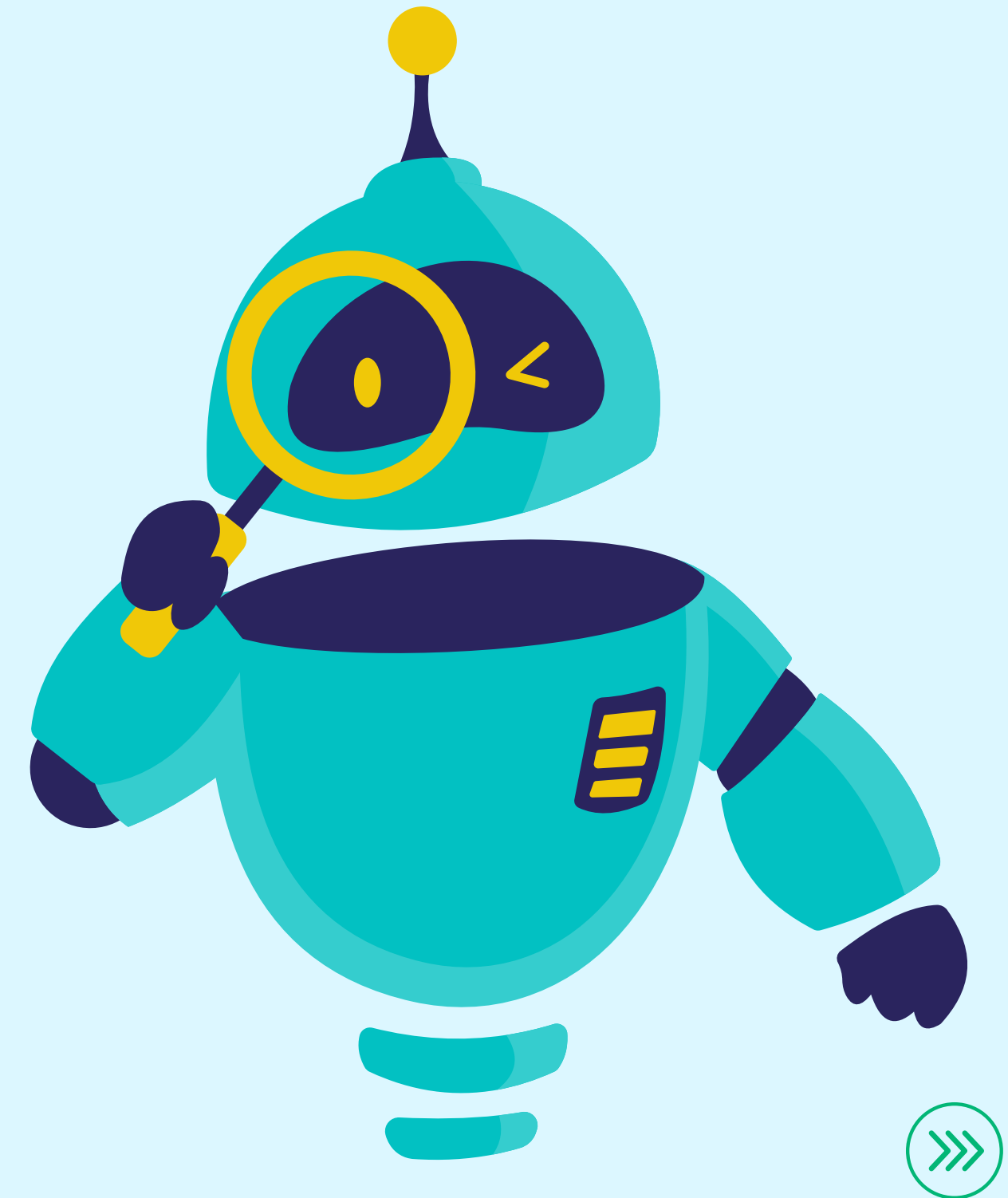


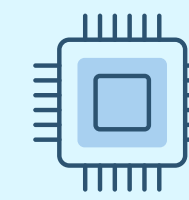
THE KEY BEHIND THE STRATEGY

Hidden Markov Model

Price Movement Predictor

Hybrid Strategy





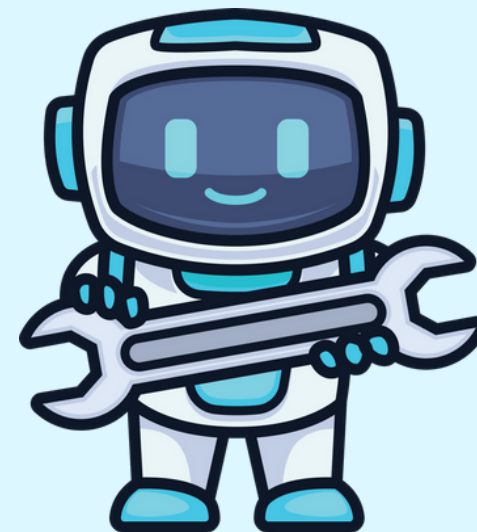
OUR VALUE PROPOSITION

HIDDEN MARKOV MODEL (HMM)

Detects stealthy shifts between bullish, bearish, and neutral phases. Helps avoid losses by adapting to hidden trends.

LSTM (TIME SERIES) + XGBOOST (FEATURE ENGINEERING):

LSTM learns sequential dependencies XGBoost ranks impactful features



HYBRID STRATEGY

Adapts actions based on regime + Machine Learning forecast.

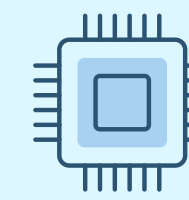
SAFE MODE

AGGRESSIVE LONG

BEARISH HMM +
SHORT-TERM LSTM
DOWNTREND

BULLISH HMM +
LSTM UPTREND





WHY IT WORKS

FUTURE-PROOF

Our hybrid system dynamically adjusts to unexpected events and Adapts to black swans.

FEDERAL RESERVE
POLICY SHIFTS

CRYPTO
CRASHES

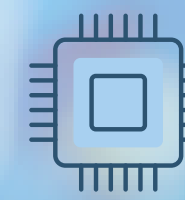
MY JOB IS HELP
TO PREVENT:

AI “THINKS IN LAYERS”

Our hybrid system mirrors human-like reasoning by combining macro-context (HMM) with micro-patterns (LSTM/XGBoost).

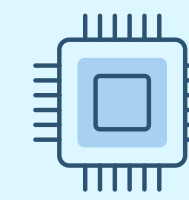
ENABLING SMARTER,
MORE CONTEXTUAL
DECISIONS





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STEP-BY-STEP METHODOLOGY



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STEP-BY-STEP METHODOLOGY

Feature Engineering

Market Regime Detection

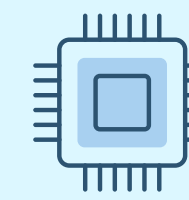
Directional Prediction

Hybrid Strategy Execution

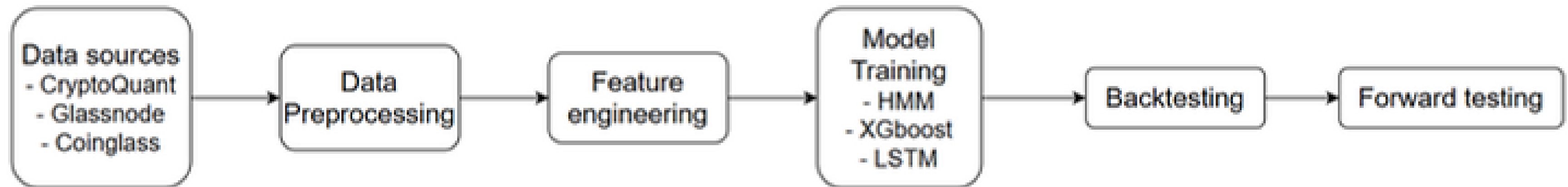
Back Testing Framework



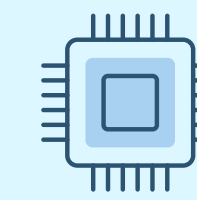
ARCHITECTURE DIAGRAM



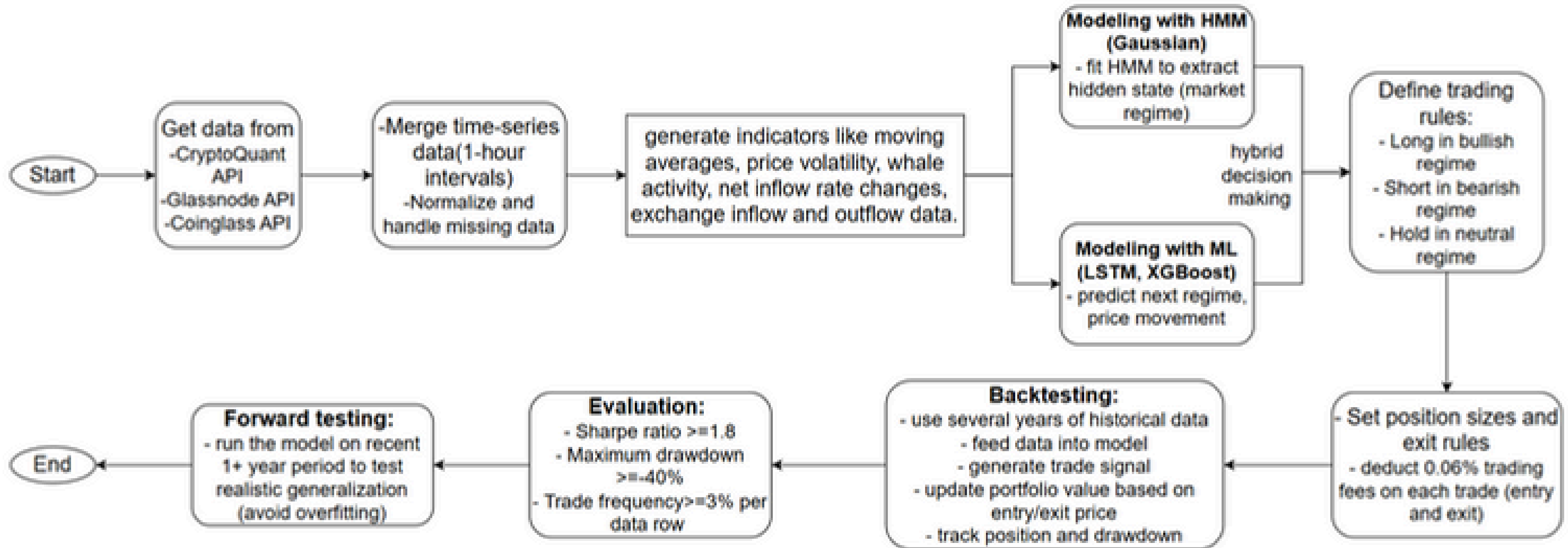
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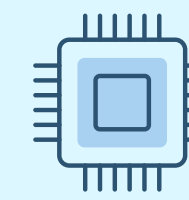


CONCEPTUAL DIAGRAM



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FEATURE ENGINEERING



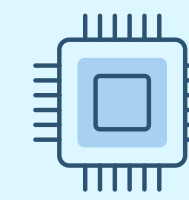
Technical Indicator:

- Moving averages (SMA 7, 14, 30 days)
- MACD with signal line and histogram
- RSI (Relative Strength Index)
- Bollinger Bands (middle, upper, lower)
- True Range and ATR (Average True Range)

Custom Features:

- Price momentum (1-day, 3-day, 7-day percentage changes)
- Rate of Change (ROC) for 5, 10, and 20 periods
- Rolling statistics (mean, standard deviation, max, min)
- Volatility features (7-day, 14-day, 30-day)
- High-Low ranges when available
- Volume-based features when available





FEATURE ENGINEERING



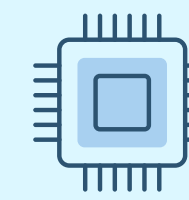
Lagged Features:

- Creates lagged price and volume features for time series modeling
- Supports configurable lookback windows

Data Normalization :

- Standardizes or min-max scales features for better model performance
- Handles missing values through forward/backward filling





MARKET REGIME DETECTION



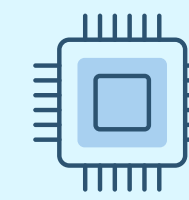
Detection Methods:

- Gaussian Mixture Models (GMM) – primary method
- K-Means clustering as alternative

Feature Selection:

- Uses volatility, rate of change, and other features to identify different market regimes
- Default features include: volatility_14d, roc_5, roc_20





MARKET REGIME DETECTION



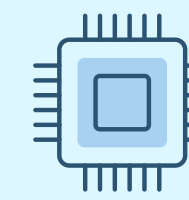
Regime Classification:

- Typically uses 2 regimes (bullish and bearish)
- Assigns a regime label to each data point
- Analyzes average returns within each regime to determine favorable/unfavorable market conditions

Integration with Trading Signals:

- Acts as a filter for the HMM model's trading signals
- Provides confirmation for buy/sell signals based on current market conditions
- Adjusts signal confidence based on regime alignment





DIRECTIONAL PREDICTION



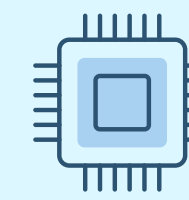
Hidden Markov Model (HMM)

- Identifies hidden market states (typically 5 states)
- Classifies states into bullish, bearish, or neutral based on return characteristics
- Calculates state-specific metrics: mean returns, Sharpe ratios, volatility
- Generates signals directly from state classifications

XGBoost Model

- Uses lagged price/feature values to predict future price movements
- Implements feature scaling for optimal performance
- Applies Savgol and Kalman filtering for signal smoothing
- Calculates derivatives for signal generation based on trend direction





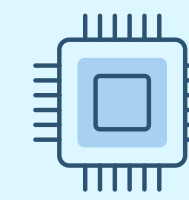
DIRECTIONAL PREDICTION



LSTM Model (Long short-term memory)

- Uses a deep neural network with LSTM layers for sequence modeling
- Creates fixed-size windows of features for prediction
- Optimizes with early stopping to prevent overfitting
- Generates trading signals based on prediction thresholds





HYBRID STRATEGY EXECUTION



Model Integration:

- Trains all models (HMM, XGBoost, LSTM) on the same data
- Configurable to exclude LSTM if desired

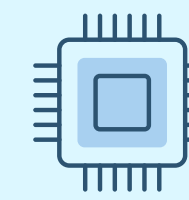
Signal Combination Strategies:

- Ensemble voting: combines signals with equal or weighted voting
- Confidence-based weighting for different market conditions
- Ensures signal alignment across all models for higher conviction trades

Execution Logic

- Position sizing based on signal strength
- Trade entry/exit based on combined signal
- Configurable to allow or restrict short selling





BACKTESTING FRAMEWORK



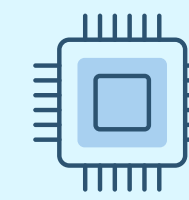
Performance Metrics:

- Total and annualized returns
- Volatility and Sharpe ratio
- Maximum drawdown
- Win rate and trade frequency
- Long/short performance analysis
- Comparison to buy-and-hold strategy

Trading Costs:

- Incorporates trading fees for realistic performance assessment
- Calculates transaction costs only when positions change





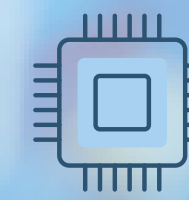
BACKTESTING FRAMEWORK



Position Management:

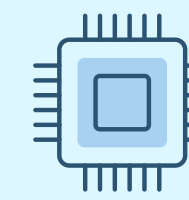
- Tracks position changes based on signals
- Adjusts returns for trading fees
- Supports both long and short positions





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TRADING RULES & SIGNAL GENERATION



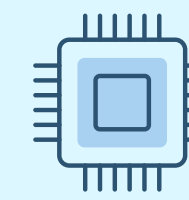
TRADING RULES & SIGNAL GENERATION

Data Selection



- Inflow/Outflow: Captures market supply-demand dynamics
- Close Price Data: Provides essential technical foundation for pattern recognition
- Exchange Whale Ratio: Detects large-scale market movements before price action
- Combined Dataset: Creates multi-dimensional view that individual indicators miss





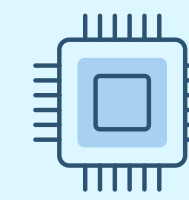
TRADING RULES & SIGNAL GENERATION

Regime-Based Execution



- Bullish Regime: Enter LONG positions when ML model predicts upward movement
- Bearish Regime: Enter SHORT positions when ML model predicts downward movement
- Neutral Regime: Remain in cash or reduce exposure to minimize unnecessary risk

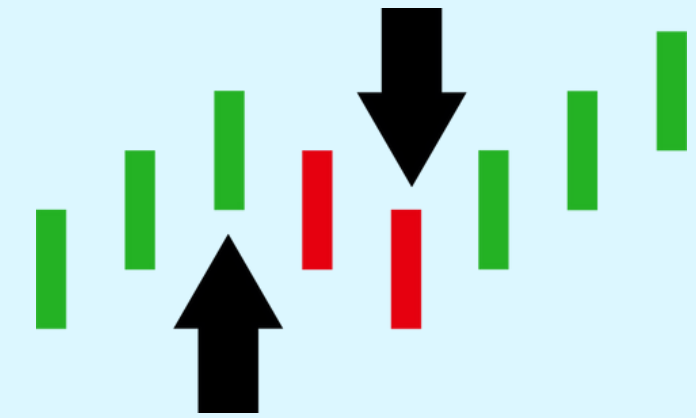




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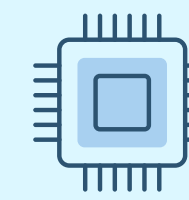
TRADING RULES & SIGNAL GENERATION

Risk Management Framework



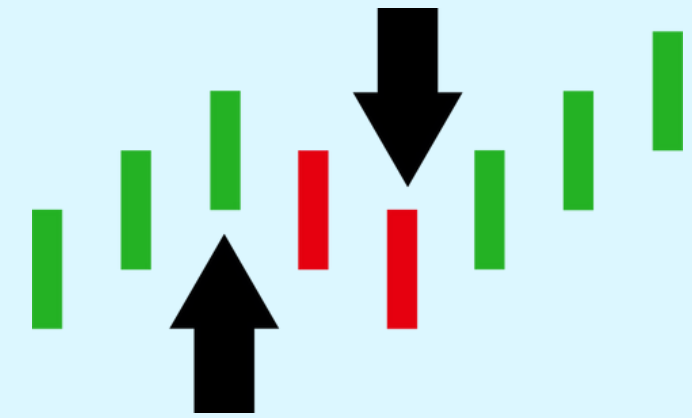
- Trading Cost Optimization: Algorithm accounts for 0.06% exchange fees
- Drawdown Control: Position sizing adjusts based on volatility to limit maximum drawdown
- Signal Filtering: Only highest-conviction signals executed to maintain quality over quantity





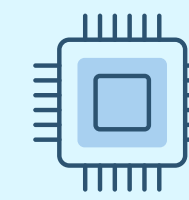
TRADING RULES & SIGNAL GENERATION

HMM Signal Generation



- **State Classification:** Classifies market into hidden states (bullish, bearish, ranging, etc.)
- **Profitability Analysis:** Identifies which states historically yield positive returns
- **Signal Rules:**
 - Long position when transitioning to profitable states
 - Short position when transitioning to unprofitable states (if shorts allowed)
 - Uses return threshold parameter to determine which states are considered profitable
 - Signal strength proportional to state probability

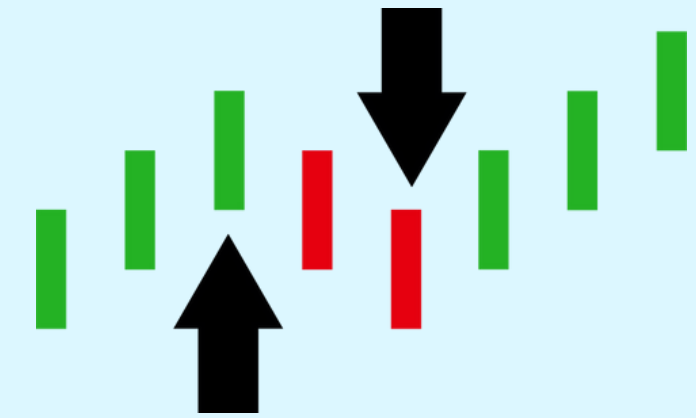




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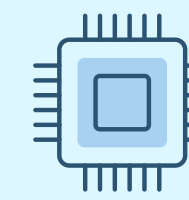
TRADING RULES & SIGNAL GENERATION

XGBoost Signal Generation



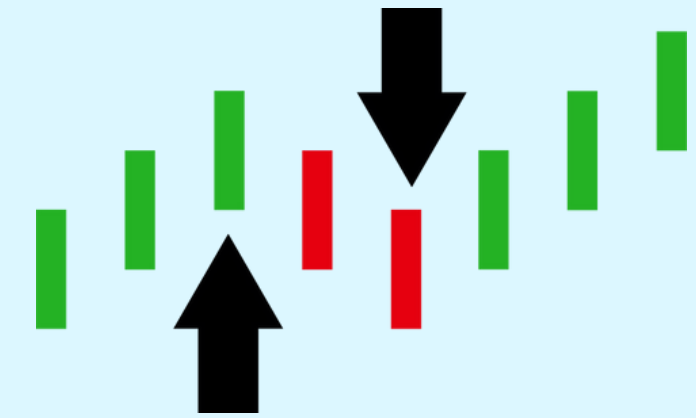
- **Direct Price Prediction:** Predicts future price movements using lagged features
- **Feature Importance:** Automatically identifies most predictive market indicators
- **Signal Rules:**
 - Long position when predicted return exceeds threshold
 - Short position when predicted return below negative threshold
 - Signal scaled based on prediction confidence





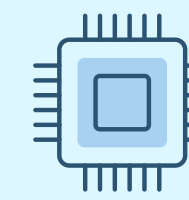
TRADING RULES & SIGNAL GENERATION

LSTM Signal Generation



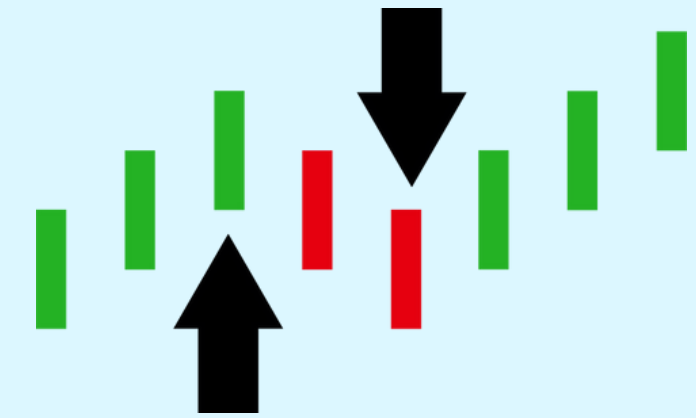
- **Sequential Pattern Recognition:** Captures complex price patterns in time windows
- **Deep Learning Based:** Learns non-linear relationships in time series data
- **Signal Rules:**
 - Generates trading signals based on predicted direction and magnitude
 - Uses sliding window of past prices to make predictions
 - Window size determines how much historical context influences predictions





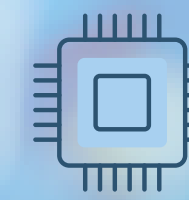
TRADING RULES & SIGNAL GENERATION

Hybrid Model Rules



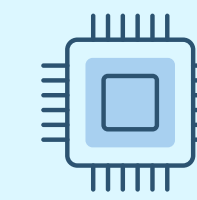
- **Ensemble Approach:** Combines signals from multiple models
- **Weighted Decision:** Final signal considers strength/confidence from each model
- **Filtering Mechanisms:**
 - Market regime detection optional (bull/bear filter)
 - Can disable short positions entirely (long-only mode)
 - Trading frequency optimization based on signal strength





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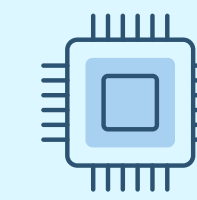
BACKTEST RESULTS



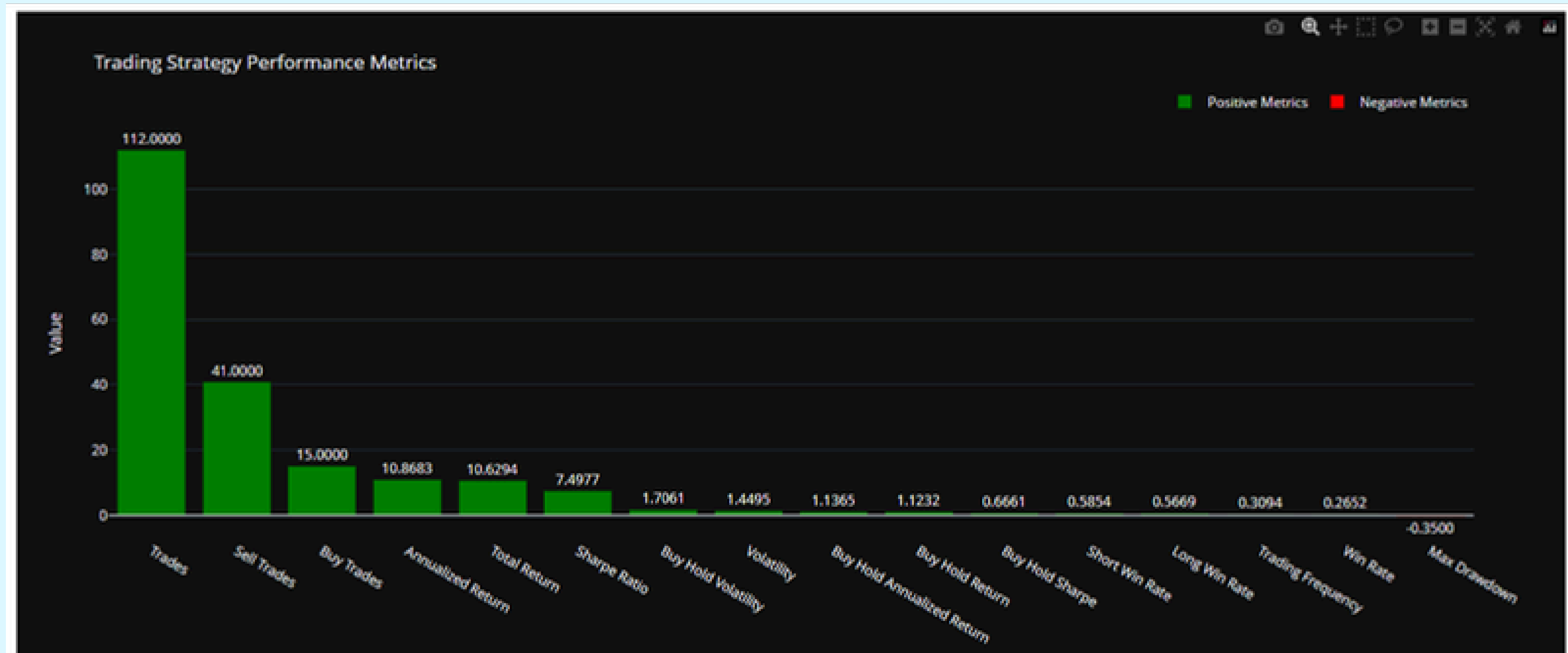
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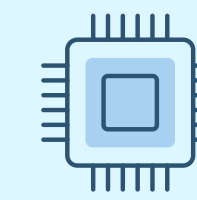
BACKTEST RESULTS





BACKTEST RESULTS

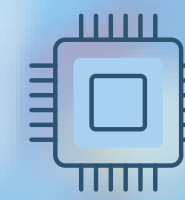




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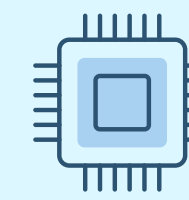
BACKTEST RESULTS





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COMPETITIVE ADVANTAGES OF ALPHA TRADING



COMPETITIVE ADVANTAGES OF ALPHA TRADING

01

SUPERIOR STATISTICAL FOUNDATION

- Hybrid model (HMM, XGBoost, LSTM) creates robust signal generation
- Kalman smoothed prediction identifies trends through noise reduction
- Derivatives analysis generates signals through threshold-based crossovers
- Multiple models compensate for individual weaknesses

02

ADAPTIVE MARKET INTELLIGENCE

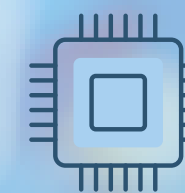
- Real-time regime classification adapts to current market conditions
- Balanced approach with effective signals in both bullish and bearish conditions
- Different strategies deployed based on identified market state
- Self-adjusting parameters optimize for current market characteristics

03

PRACTICAL IMPLEMENTATION

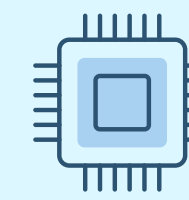
- Fast execution (sub-microsecond speed)
- Eliminates emotional trading decisions
- Generates passive, automated income
- Scalable across multiple cryptocurrencies





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WHAT WE NEED



RESULTS

DATA PREPARATION

- Data collection, cleaning, and preprocessing
- Ensuring data quality and reliability
- Data labeling and annotation

MODEL DEVELOPMENT

- Selection of appropriate machine learning algorithms
- Model architecture and hyperparameter tuning
- Training process and validation

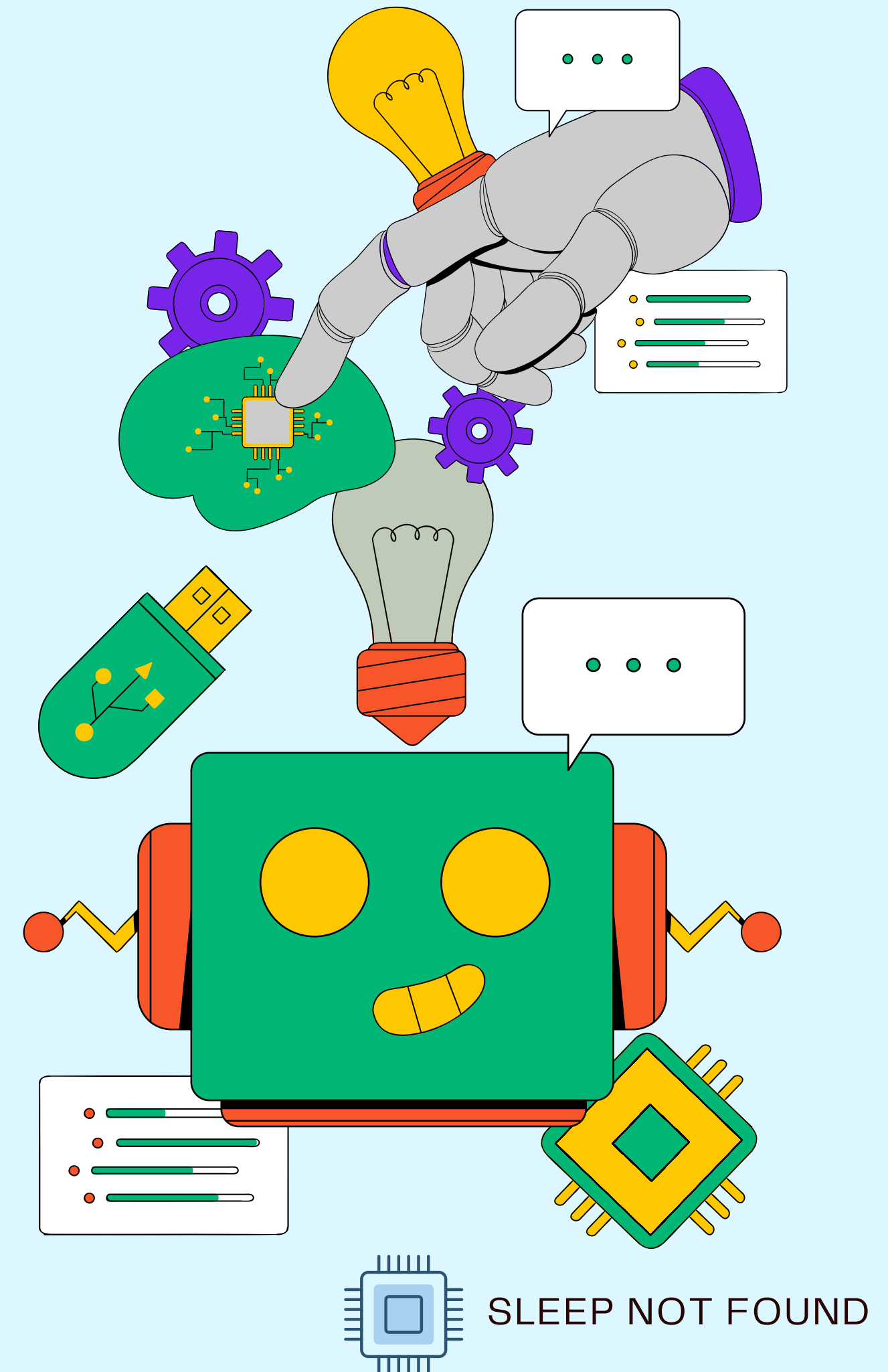
MODEL EVALUATION

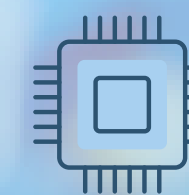
- Cross-validation and testing on validation datasets
- Assessing model performance, including accuracy, precision and recall



CONCLUSION

- Successfully implemented a comprehensive and adaptive cryptocurrency trading system that integrates HMM, XGBoost, and LSTM neural networks.
- It accurately detects market regimes, leverages both technical and on-chain features, and dynamically adjusts trading strategies in response to changing market conditions.
- The model achieved significantly higher returns and Sharpe Ratio compared to a traditional buy-and-hold approach, with reduced drawdowns and consistent performance across bull and bear markets, demonstrating its robustness and real-world applicability.





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THANK YOU